

재료공학개론 I (이,공) (MSE305a-01)

학기	2026년 1학기
담당교수	이주혁
이수구분	교선평수
학점	3.0
강의시간	월09:00-10:30(E7 - L20), 수09:00-10:30(E7 - L20)
강의실	E6 411 (교수실)
소속	기초학부

교과개요

The purpose of this course is to provide a conceptual framework for understanding the behavior of engineering (solid) materials by emphasizing important relationships between internal structure and properties. It attempts to present a general picture of the nature of materials and the mechanisms that act upon, modify, and control their properties. The subject matter is meant to provide prospective scientists with sufficient background and understanding to appreciate existing materials and to exploit new materials development effectively in energy-related research and engineering. This course has three main parts. The first part treats the internal structures of perfect solids including metals, ceramics, and polymers. The second part deals with defects in solids as well as kinetic problems of diffusion, and connected mechanical properties. The third part covers basic concept of equilibrium in solids (phase diagram), phase transformations, and structure control. We also have optional chapters of material properties of solids including electrical properties showing the relation to the structure.

교과목표

Gain basic knowledge of material science and engineering by acquiring basic knowledge of atomic structure and bonding, defects, phase equilibrium, phase changes, and types of materials

주차별 강의계획

- 1주차:Introduction
- 2주차:Atomic Structure and Bonding
- 3주차:Atomic Structure and Bonding
- 4주차:Introduction to Crystallography
- 5주차:The Structure of Crystalline Solids
- 6주차:The Structure of Crystalline Solids
- 7주차:Structures of Polymers
- 8주차:Midterm exam
- 9주차:Defects in Solids
- 10주차:Diffusion in Solids
- 11주차:Diffusion in Solids
- 12주차:Phase Diagram
- 13주차:Phase Transformation
- 14주차:Synthesis, Fabrication, and Processing of Materials
- 15주차:Composites
- 16주차:Final exam

수업방법

Theory and practice

학습윤리

Trust the value of your own intellect. Honesty is the foundation of good academic work.

교과정책(방침)

Allow use of laptop during class.

준비물 및 기타

Calculator